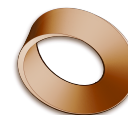


L3 – Kinetic Energy

P1 Energy • AQA Combined Science: Trilogy 8464



Physics 10 P and 10 R • Mr. Rawson

10 P	Wed 2 Sep, P2, 09:55–10:45	Next	Tue 8 Sep — 6 days
10 R	Thu 3 Sep, P2, 09:55–10:45	Next	Thu 10 Sep — 7 days
Spec	P1.1.2 — aP1-5, aP1-6	Lesson	3 of 11
Equation	$E_k = \frac{1}{2}mv^2$ — RECALL . Not on the equation sheet. Say that out loud.		
To print	Per class (19 for 10 P, 11 for 10 R): the L3 handout, the homework sheet, <i>and</i> Skills/skills-unit-conversion-handout.pdf — issued as a reference they are expected to keep, not taught.		
Homework	Six calculations, graduated, the last two requiring rearrangement for v . <i>Both classes then have a week away</i> — this is the only physics they will touch.		

By the end of the lesson students can:

- recall $E_k = \frac{1}{2}mv^2$ and state the unit of every quantity in it;
- calculate the kinetic store of a moving object, and rearrange to find v ;
- say what happens to the kinetic store when speed doubles, *and why*.

Timing

Time	Phase	What happens
0–5	Do Now	Quickfire: store or pathway? Six items on the board. This is L2's hinge — if it did not land, it costs you the lesson.
5–13	The equation	$E_k = \frac{1}{2}mv^2$. Units first: kg, m/s, J. Say it is a recall equation; have them write it inside the front cover.
13–20	Worked example	The cyclist, on the board, you narrating. One example only — the km/h case is now handled properly at 35.
20–32	Handout Q1 a–c	Circulate hunting for one specific error: v not squared. Part (c) is the same car at double the speed, so the $\times 4$ arrives in their own working rather than from you.
32–38	Q2 and Q3	The table, then say it out loud: double the speed, four times the store. It returns in P5 stopping distances.
38–41	Conversion sheet	Hand it out and set the expectation rather than teaching it: this is a skill they are expected to have and to keep. Do not work through it. <i>Do</i> put the one line $72 \text{ km/h} \rightarrow 20 \text{ m/s}$ on the board — Q4 depends on it, and three minutes now is cheaper than a wrong answer you do not see until October.
41–50	Handout Q4 and Q5	The 72 km/h car, then the rearrangement. Set homework.

Questions to ask

1. A car doubles its speed. What happens to its kinetic store, and why not just double?
2. A lorry at 10 m/s or a car at 20 m/s — which has the bigger kinetic store? What else do you need to know?
3. Why is 60 mph not simply “twice as dangerous” as 30?
4. Which of the two quantities in this equation would you rather halve, and why?
5. Where did the kinetic store come from, and where will it go?

Common misconceptions

- **v not squared.** The single most common arithmetic error in P1. Circulate for it rather than waiting to mark it.
- $(\frac{1}{2}mv)^2$. Squaring the whole expression rather than only v . Show both on the board and let them see the difference.
- **Speed in km/h or mph.** The equation needs m/s. Build the conversion into example 2 deliberately.
- **“Heavier means more kinetic energy.”** True only at equal speed. The squared term usually wins — question 2 above exists to expose this.

Differentiation

Support: a substitution scaffold — boxes labelled m , v , v^2 , then E_k — so the squaring step is a physical box that cannot be skipped.

Extend: rearrange for v before the class reaches it, then ask for the speed a 1500 kg car needs to match the kinetic store of a lorry.

After the lesson

- **10 P runs first this time** — the reverse of L2. Adjust before Thursday.
- **Both classes now vanish for about a week.** Make sure the homework actually went home, and that they can recall the equation without the book.
- L4 is gravitational and elastic potential energy. If the squaring error was widespread, open L4 with it rather than moving on.
- **The conversion sheet is a keeper.** Tell them to file it, not bin it — it is needed again in P5 Forces and in every speed calculation this year.
- **Q4 is your diagnostic.** It is the only place the expectation gets tested. If a fair number get 259 200 instead of 20, the skill is not there and pretending otherwise costs you all year — take ten minutes in L4 rather than hoping.
- Mark both Class Log entries **Taught**.